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Enhanced blood coagulation and fibrinolysis in mice lacking histidine-rich glycoprotein (HRG)

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To cite this article: Tsuchida-Straeten N, Ensslen S, Schäfer C, Wöltje M, Denecke B, Moser M, Gräber S, Wakabayashi S, Koide T, Jahnén-Dechent W. Enhanced blood coagulation and fibrinolysis in mice lacking histidine-rich glycoprotein (HRG). *J Thromb Haemost* 2005; 3: 865–72.

See also Borza D-B. Life without histidine-rich glycoprotein: modulation of the hemostatic balance revisited. This issue, pp 863–4.

Summary. Histidine-rich glycoprotein (HRG) is a serum protein belonging to the cystatin superfamily. HRG may play a regulatory role in hemostasis and innate immunity. However, this role is uncertain because of a lack of rigorous testing in an animal model. We generated mice lacking the translation start point of exon 1 of the *Hrg* gene, effectively resulting in a null mutation (*Hrg*^{−/−}). The mice were viable and fertile but had no HRG in their blood. Antithrombin activity in the plasma of *Hrg*^{−/−} mice was higher than in the plasma of heterozygous *Hrg*^{+/-} or wild-type *Hrg*^{+/+} mice. The prothrombin time was shorter in *Hrg*^{−/−} mice than in *Hrg*^{+/-} and *Hrg*^{+/+} mice. Bleeding time after tail tip amputation in *Hrg*^{−/−} mice was shorter than in *Hrg*^{+/-} mice. The spontaneous fibrinolytic activity in clotted blood of *Hrg*^{−/−} mice was higher than in *Hrg*^{+/-} mice. These findings suggest that HRG plays a role as both an anticoagulant and an antifibrinolytic modifier, and may regulate platelet function *in vivo*.

Keywords: blood proteins, glycoproteins, hemostasis, HRG, knockout mice.

Introduction

Human histidine-rich glycoprotein (HRG) is a single-chain glycoprotein with a molecular weight of 67 000; it is composed of 507 amino acid residues. HRG belongs to the cystatin superfamily of proteins like fetuins and kininogens [1–3]. The

genes for fetuin-A, fetuin-B, HRG and kininogen cluster head to tail on chromosome 3 in the human genome and on chromosome 16 in mouse. The *Hrg* gene includes seven exons and six introns. The N-terminal region of the protein consists of two cystatin-like domains D1 and D2 which are each encoded by three exons, exons I to III and exons IV to VI, respectively. Exon VII encodes for a large C-terminal domain, whose N-terminal region is very rich in proline and histidine residues. HRG is exclusively synthesized in the liver, but it becomes associated with platelets and megakaryocytes and can be released after thrombin stimulation [4]. Mature HRG circulates in the blood at a relatively high concentration of up to 150 mg L^{−1} (~2 μmol L^{−1}).

HRG binds to heparin [5,6], plasminogen [7,8] and fibrinogen [9] suggesting a role in hemostasis. Several individuals with a congenital deficiency of HRG were discovered in three families with thrombophilia [10–12] corroborating such a role. Molecular binding of HRG to multiple ligands is mediated through specific binding domains, which bind independent of each other [13]. A low-affinity binding site for the cell surface glycosaminoglycan heparan sulfate was recently mapped to the N-terminal tandem cystatin domains (N1N2) [14]. Cell binding of HRG was enhanced by Zn²⁺ in an indirect fashion by binding to the histidine-proline-rich domain. This binding behavior can greatly influence HRG bioactivity at sites of local changes in pH value and Zn²⁺ concentration [15], suggesting that HRG may be a tunable adaptor protein for several ligands involved in inflammation and tissue remodelling. Such a mechanism was indeed experimentally verified for HRG binding of plasminogen [16]. The authors proposed that under conditions of low pH and elevated free Zn²⁺, HRG acts as a high-affinity receptor for plasminogen thus enhancing the migratory potential of cells.

In addition, HRG binds and transports divalent metals [17,18], to complement components and IgG, and inhibits T-cell proliferation [19–21]. Recently, it was reported that HRG has potent antiangiogenic activity [22,23]. Despite these abundant biological properties reported for HRG, the physiological

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Received 28 September 2004, accepted 16 December 2004

function of HRG has not been firmly established. To explore the physiological function of HRG, we cloned the mouse HRG gene and produced, by homologous recombination, mice lacking HRG.

Materials and methods

All animal experiments in this study were performed complying with the German Animal Protection Law (Tierschutzgesetz) following procedures reviewed and approved by institutional and state animal protection boards.

Targeting vector and homologous recombination

To construct a targeting vector for the *Hrg* gene, we first screened a mouse 129/SvJ genomic library with a 1.1-kilobase (kb) *EcoRI* fragment of rat HRG cDNA. The identity of the gene was confirmed by restriction mapping with rat and mouse *Hrg* cDNAs and by sequencing. A targeting vector was constructed following published protocols [24]. A 1.9-kb *SalI-XbaI* DNA fragment including a part of exon I of the mouse *Hrg* gene served as 5'-homologous region and a 7.8-kb *BamHI* genomic fragment of mouse *Hrg*, which contains exons II to VII served as a 3'-homologous region. Thus, the targeting vector replaced a 1.8-kb *XbaI-BamHI* fragment comprising a part of exon I including the translation start point with a neomycin cassette for positive clone selection (Fig. 1). Flanking TK cassettes for negative selection were also contained in the vector.

129/Ola embryonic stem cells (passage 10) were grown on a mouse embryonic fibroblast feeder layer and electroporated with 60 µg of *PvuI*-linearized replacement vector DNA following established protocols. After positive-negative selection using G418 (400 µg mL⁻¹ final concentration) and ganciclovir

(2 µmol L⁻¹ final concentration), 144 resistant clones were isolated, cryo-conserved and used for DNA preparation. DNA samples from individual ES cell clones digested with *ScaI* or *HindIII* were probed with the 0.6-kb *SmaI* fragment located immediately upstream of the 1.9-kb *SalI-XbaI* fragment used in the replacement vector. This probe detected a 10.4-kb *ScaI* fragment and a 4.6-kb *HindIII* fragment in wild-type DNA or a 6.5-kb *ScaI* fragment and a 8.4 kb *HindIII* fragment in the correctly targeted allele (see Fig. 1).

Two independent ES cell clones (clone 33 and clone 24) yielded fertile chimeras upon blastocyst injection. One clone 33 chimeric female and one clone 24 chimeric male transmitted the targeted *Hrg* locus through the germline as demonstrated by Southern blot analysis of tail DNA from F₁ offspring.

Genotypes were determined by Southern blot hybridization analysis as described for ES cells or by polymerase chain reaction (PCR). PCR analysis was carried out using primers P1 (5'-TTGAATGATTCTAATGTGAAGGC-3'), P2 (5'-GGT CCTCATTTCCGCATTCCGTGGA-3') and P3 (5'-TTGT GTAGCGCAAG TGCCAGCG-3') covering the positions indicated in Fig. 1.

Immunoblots were performed with 0.2 µL of serum and rabbit antiserum against human HRG, which cross-reacts with mouse HRG as described [25].

Hematological analysis

Blood was collected by cardiac puncture and serum was obtained by centrifugation. Serum chemistry and hemograms were performed using a Vitros 250 analyzer (Orthochemical Diagnostics, Necher Giehlind, Germany) and a Celltek-M (Nihon Konden; Johnson & Johnson, Bad Hamburg, Germany) analyzer, respectively. For coagulation assays blood was anticoagulated with sodium citrate (9 : 1 v/v with 3.8% sodium citrate). Antithrombin activity was determined using the COAMATIC[®] Antithrombin kit (Chromogenix, Orangeburg, NY, USA) and control plasma N (Dade Behring, Marburg, Germany).

Fibrinogen was determined using commercial kit 886 (Sigma, Taufkirchen, Germany). Prothrombin time was measured using a KC10A coagulometer (Amelung, Lemgo, Germany). Human HRG was purified from human plasma as described previously [26].

Bleeding time

The tails of 2–4-month-old anesthetized mice (100 mg ketamine and 5 mg xylazine per kg intraperitoneally) were prewarmed for 5 min at 37 °C in a water bath, and a 5-mm segment of tail tip was amputated with a razor blade. The bleeding time was taken until the bleeding stopped.

Whole blood clot lysis

Briefly, 100 µL of fresh blood collected by vena cava puncture was diluted 10-fold with saline containing ~10 000 counts per

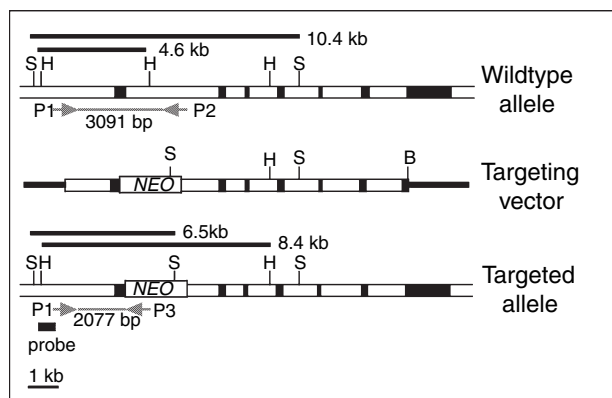


Fig. 1. Structure of the mouse *Hrg* gene. The mouse *Hrg* gene and its partial restriction map are schematically drawn as the wild-type allele (top). Exons are represented by black boxes. Restriction enzymes are *BamHI* (B), *HindIII* (H), and *ScaI* (S). The targeting vector is drawn in the middle. The 3'-portion of exon I and the 5'-half of intron A were replaced by the neomycin cassette in the targeted *Hrg* gene (bottom) effectively removing the translation start. The position of PCR primers (P1, P2, and P3) and the size of the expected PCR products are shown below the genes. Restriction fragments digested by *ScaI* or *HindIII* restriction enzymes are indicated with their respective sizes in kb probed by a 0.6-kb *SmaI* fragment.

min human ^{125}I -labeled fibrinogen (Amersham, Buckinghamshire, UK) and was coagulated at 37 °C with 8 μL of bovine thrombin (75 U mL^{-1}) (Sigma), and 10 μL of 0.75 mol L^{-1} CaCl_2 solution [27]. At the times indicated the ^{125}I content of 50- μL aliquots of supernatant was measured using a gamma-counter (model LB 951 G; Berthold, Bad Wildbad, Germany). The release of ^{125}I -labeled fibrin degradation peptides was analyzed by sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) and autoradiography using 10 μL of lyzed clot supernatant and NuPAGETM acrylamide gels (Novex, Frankfurt, Germany).

Induction of skin wounds

Mice were anesthetized, shaved dorsolaterally and swabbed with 70% ethanol. A full thickness wound was inflicted using an 8-mm dermal biopsy punch. Wound areas were photographed daily and the wound area was measured histomorphometrically using National Institutes of Health IMAGE software.

Results

Structure of the *Hrg* gene locus

Figure 1 shows the structure of the mouse *Hrg* gene locus. The complete nucleotide sequence of 13 591 bases (AB055898) was determined by sequencing. The alignment with the cDNA sequence demonstrated that the gene for mouse *Hrg* was composed of six small exons I through VI (81–225 bp), and one large exon VII (905 bp). The size of introns varied from 677 bp to 2315 bp. This exon–intron structure was very similar to that of the human gene [28] where seven exons were likewise identified. The structural organization into two groups of three exons coding for a cystatin-like protein domain each, and one additional larger exon coding for a third protein domain specific for the respective protein is also present in the related fetuin family proteins, Fetuin-A/*Ahsg* and Fetuin-B/*Fetub* [24,29]. The genes encoding these proteins are neighbors on chromosome 16 in the mouse genome and therefore originated by gene duplication. A similar arrangement is conserved in the human genome.

Generation of HRG null mutant mice (*Hrg*^{−/−})

A targeting vector for homologous recombination was constructed as shown in Fig. 1. Upon genomic Southern hybridization of DNA isolated from 144 neomycin-resistant cell clones three correctly targeted embryonic stem cell clones with the predicted restriction patterns (Fig. 1) were identified. Two independent ES cell clones were successfully used to derive chimeric mice after blastocyst injection and embryo transfer. Genomic DNA analysis of F₁-progeny of these mice indicated the presence of one wild-type allele and one null allele derived by homologous recombination at the *Hrg* locus. Mice of all three *Hrg* genotypes were born from crosses of heterozygous

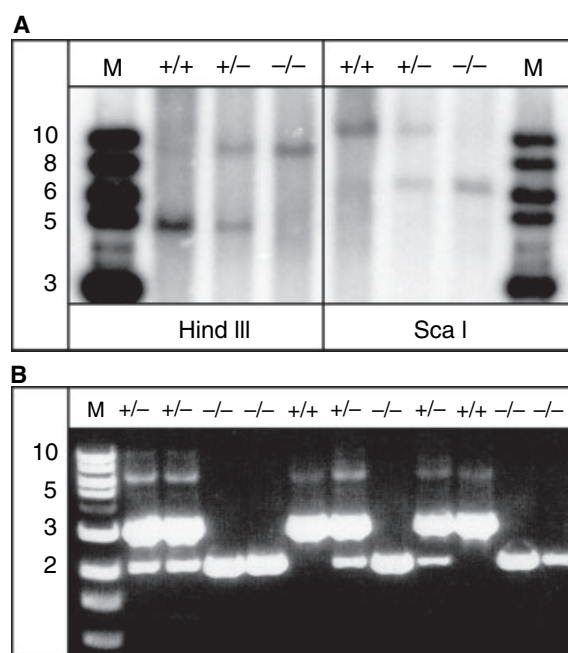


Fig. 2. Confirmation of *Hrg* gene knockout at the level of genomic DNA. (A) Southern blot analysis of genomic DNA isolated from tail biopsies of *Hrg*^{+/+}, *Hrg*^{+/-} and *Hrg*^{-/-} mice. The genomic DNA probe indicated in Fig. 1 detected a 4.6-kb *Hind* III fragment and a 10.4-kb *Sca* I fragment in the wild-type allele and an 8.4-kb *Hind* III fragment and a 6.5-kb *Sca* I fragment in the mutant allele, respectively, indicating correct gene targeting as outlined in Fig. 1. PCR analysis of the progeny of a typical heterozygous *Hrg*^{+/-} mating. (B) PCR detection of the disrupted allele was accomplished with forward primer 1 (P1, see Fig. 1) derived from the 5' non-coding flanking region of the *Hrg* gene and with reverse primer 3 (P3), located in the neomycin cassette. Thus the PCR generates an amplicon size of 2077 bp for the mutated allele. The PCR of the wild-type allele with forward primer 1 and reverse primer 2 (P2) located within the deleted region of the *Hrg* gene results in an amplicon size of 3091 bp. Note that all three genotypes were represented in one progeny. DNA marker (M) size in kb is indicated on the left.

F₁ offspring (Figs 2A,B). Ten matings yielded a combined progeny of 98 mice comprising 30 *Hrg*^{+/+}, 34 *Hrg*^{+/-}, and 34 *Hrg*^{-/-} mice illustrating that HRG-deficient mice are viable. Progeny derived from different injection experiments were phenotypically indistinguishable. The *Hrg* null mutation in its homozygous state was maintained on a mixed 129/Ola, C57BL/6 background. For further reference the mutant strain is named 129/B6-*Hrg*^{tm1Wja} according to ILAR (Institute of Laboratory Animal Resources) rules.

Analysis of HRG-deficient mice (129/B6-*Hrg*^{tm1Wja})

Homozygous HRG-deficient mice were viable and fertile. Spontaneous mortality during the first 3 months was normal at 2.6% (seven out of a total of 268 animals). We performed immunoblot analysis to confirm the HRG deficiency in serum. Figure 3 shows that two protein bands characteristic of HRG (80 kDa) and its major proteolytic degradation product were detected in the serum of wild-type and heterozygous *Hrg*^{+/-} mice, but no protein band was detected in serum from *Hrg*^{-/-}

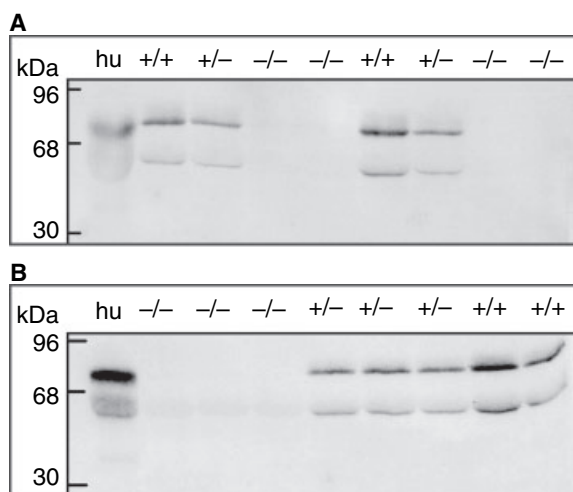


Fig. 3. Immunoblot analysis of serum *Hrg* expression. Serum (0.2 μ L) from progeny of a heterozygous F₂ cross of clone 24 (A) or clone 33 (B) was separated by reducing SDS-PAGE and electrotransferred onto nitrocellulose. HRG protein was detected by immunological staining using rabbit polyclonal antibody raised against human HRG, peroxidase-labeled secondary antibody and enhanced chemiluminescence detection. +/+, wild-type; +/-, heterozygous; -/-, homozygous HRG-deficient; hu, human serum. Note that no HRG was detected in *Hrg*^{-/-} serum and about half the amount of HRG was detected in serum from *Hrg*^{+/-} mice when compared to wild-type *Hrg*^{+/+} mice.

mice. Roughly half the amount of HRG protein could be detected in serum from *Hrg*^{+/-} mice compared to *Hrg*^{+/+} mice. This finding indicates that both copies of the *Hrg* gene are independently expressed. We obtained identical results when we analyzed HRG levels in serum from the offspring of ES cell clones 24 or 33 (Fig. 3), indicating that both strains represented a true HRG null mutation at the serum protein level. Strain 33 was analyzed in more detail.

Hematological analysis

Routine hematology revealed similar blood cell counts for *Hrg*^{-/-} mice and for wild-type mice, but monocytes were 2.5-fold more abundant in HRG-deficient than in wild-type mice ($5.3 \pm 3.0\%$ vs. $2.1 \pm 2.1\%$, $n = 8$, $P = 0.018$). Serum activity GOT/AST, GPT/ALT, GGT, blood glucose, triglycerides, urea, creatinine, and electrolytes were all normal in HRG-deficient and wild-type mice suggesting that liver, heart and kidney metabolism were uncompromised by the lack of HRG (data not shown).

Blood coagulation

We estimated antithrombin activity using a chromogenic assay measuring the residual activity of excess factor Xa in the presence of a fixed amount of heparin. Commercial human control plasma N activity was set to 100% in this assay. Plasma samples obtained from wild-type *Hrg*^{+/+} mice scored 107% compared to human control plasma N. Plasma obtained from heterozygous *Hrg*^{+/-} mice had an elevated antithrombin

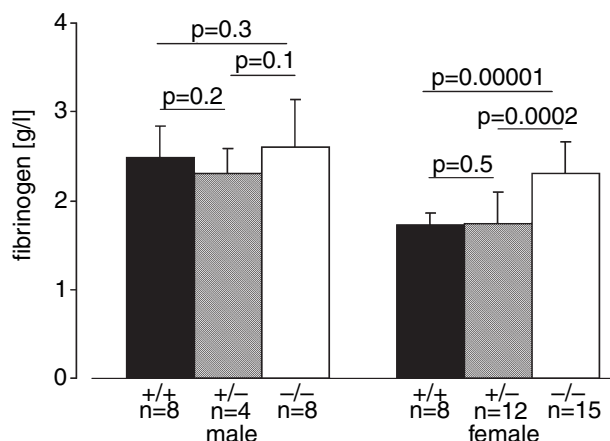


Fig. 4. Fibrinogen concentration in plasma of wild-type and HRG-deficient mice. +/+, wild-type; +/-, heterozygous; -/-, homozygous HRG-deficient. Values represent the mean $\pm$ SD of 4–15 individual mice as detailed in the figure. The *P*-values were calculated with Student's *t*-test.

activity of 116% and HRG-deficient plasma from *Hrg*^{-/-} mice had an elevated antithrombin activity of 125%. The difference in antithrombin activity between *Hrg*^{+/+} and *Hrg*^{-/-} mice was statistically highly significant ($n = 10$ each, $P = 0.001$, data not shown).

We measured fibrinogen in plasma samples from male and female mice (Fig. 4). The amount of fibrinogen was higher in plasma from males than from females. This sex-specific difference was 47% in wild-type mice (2.5 ± 0.4 g L⁻¹ male, 1.7 ± 0.1 g L⁻¹ female) to 35% in heterozygous *Hrg*^{+/-} mice (2.3 ± 0.3 g L⁻¹ male, 1.7 ± 0.4 g L⁻¹ female) and 13% in *Hrg*^{-/-} mice (2.6 ± 0.5 g L⁻¹ male, 2.3 ± 0.4 g L⁻¹ female). Plasma fibrinogen was increased in HRG-deficient mice compared to wild-type and *Hrg*^{+/-} mice. This difference was most pronounced in the plasma of HRG-deficient female mice, containing 35% more fibrinogen (2.3 ± 0.4 g L⁻¹) than plasma from heterozygous and wild-type females (both 1.7 ± 0.1 g L⁻¹).

Figure 5(A) illustrates that the prothrombin time was significantly shorter in the plasma of HRG-deficient mice (lane 4) than in human plasma (lane 1), fetuin-A-deficient (lane 2) or wild-type mouse plasma (lane 3), suggesting that HRG deficiency was associated with faster extrinsic blood clotting. This accelerated prothrombin time was specific for the HRG deficiency, because deficiency in fetuin-A, a genetically – but not functionally – related plasma protein, was not associated with shortened prothrombin time. In addition, reconstitution with purified human HRG of HRG-deficient mouse plasma restored the prothrombin time to wild-type levels (Fig. 5, lanes 4–7). Figure 5(B) depicts the immunoblot analysis of the identical plasma samples analyzed in Fig. 5(A), confirming that the clotting time increased with the amount of HRG in the plasma samples.

Next we measured the bleeding time after amputation of a 5-mm length of tail tip as an *in vivo* parameter of combined plasma and platelet mediated blood coagulation. The bleeding

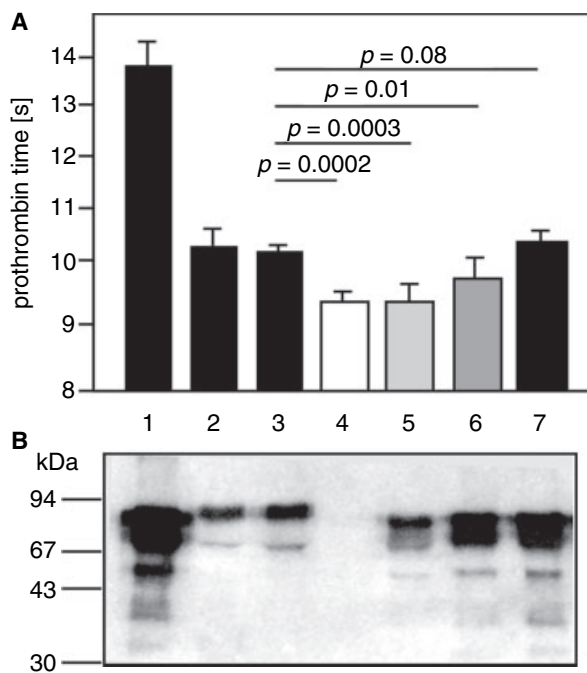


Fig. 5. Prothrombin time measurement in human and mouse plasma. (A) Prothrombin time was measured using a coagulometer. (B) Immunoblot analysis of the plasma samples used in (A). Lane 1, human plasma; lane 2, mouse fetuin-A-deficient plasma; lane 3, wild-type plasma; lane 4, HRG-deficient plasma; lanes 5–7, HRG-deficient plasma with 10, 30 and 60 $\mu\text{g mL}^{-1}$ human HRG added, respectively. Note that the reconstitution with human HRG (60 $\mu\text{g mL}^{-1}$) of HRG-deficient mouse plasma restored normal prothrombin time dose-dependently. Values represent the mean $\pm$ SD of four individual mice. The *P*-values were calculated with Student's *t*-test.

times were highly variable, ranging from 0.5 to 5.5 min (Fig. 6). Nevertheless, the bleeding time was significantly shorter in *Hrg*^{-/-} mice compared to *Hrg*^{+/+} mice (median $\pm$ SEM, 50 $\pm$ 8 s, *n* = 14 vs. 62 $\pm$ 22 s, *n* = 23, *P* < 0.01). *Hrg*^{+/-} mice showed bleeding times similar to wild-type mice (63 $\pm$ 10 s, *n* = 27).

Taken together the results from measurements of antithrombin activity, prothrombin time and bleeding time suggest a mild, yet statistically significant, anticoagulant action of plasma HRG.

Next we estimated the fibrinolytic activity in vena cava blood clotted by the addition of bovine thrombin and ¹²⁵I-labeled fibrinogen. Figure 7(A) illustrates that spontaneous clot lysis was faster in HRG-deficient mice than in wild-type mice. Fifteen minutes after initiation of blood clotting by the addition of thrombin we observed 4.36 $\pm$ 0.24% release of input radioactive label into the soluble supernatant in blood from *Hrg*^{-/-} mice vs. 1.30 $\pm$ 0.10% in wild-type mice. The accumulated release of ¹²⁵I label in *Hrg*^{-/-} and wild-type mice ranged from 8.74 $\pm$ 0.31% vs. 2.66 $\pm$ 0.23% at 30 min, to 44.32 $\pm$ 2.74% vs. 14.90 $\pm$ 0.23% at 22.5 h. To visualize the fibrin degradation peptides liberated during fibrinolysis we analyzed the lyzed clot supernatant by SDS-PAGE and

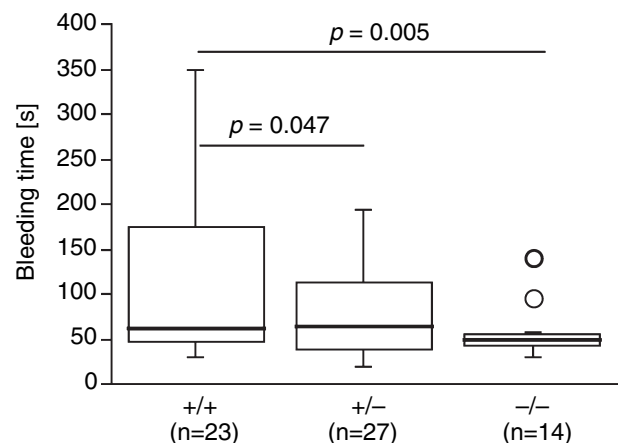


Fig. 6. Bleeding time. Tail tips were clipped and the bleeding time was determined as described in the Materials and methods. Data are displayed in a box plot of the indicated number of individual measurements for each genotype. Each box encloses 50% of the data with the median value of the variable displayed as a line. The top and bottom of the box mark the limits of $\pm$ 25% of the variable population. The lines extending from the top and bottom of each box mark the minimum and maximum values within the data set that fall within an acceptable range. Any value outside this range, called an outlier, is displayed as an individual point. Outliers are included in all calculations. *P*-values of Student's *t*-tests are indicated at the top of the columns.

autoradiography (Fig. 7B). In full accordance with the faster clot lysis in *Hrg*^{-/-} mice determined by radioactive measurement of clot lysis supernatants, higher amounts of radioactive ¹²⁵I-labeled fibrin degradation peptides were consistently detected in lyzed clot supernatants of HRG-deficient mouse blood (Fig. 7B, lane 5–8) than in lyzed clot supernatants from wild-type mouse blood (lane 1–4). Taken together these results suggest that HRG may act as an antifibrinolytic *in vivo*.

Skin wound model

To evaluate if the observed differences in both coagulation and fibrinolysis would influence the healing of an external wound, which involves both thrombosis and fibrinolysis, we inflicted full thickness skin wounds to the back of *Hrg*^{-/-} and wild-type mice. One day after excision of skin biopsies the wounds had shrunk to about 50% of the original wounded area in both wild-type and *Hrg*^{-/-} mice. The wounds were typically covered by a dry eschar within 2 days (not shown). A detailed morphometric analysis of the wound healing was performed using electronic images of the wounds. Large variation of wound areas existed within experimental groups during the wound contraction period (day 1–4) regardless of their genotype. These differences leveled out during the first week. We did not observe statistically significant differences in the gross morphology (not shown) and timing of wound healing between wild-type and *Hrg*^{-/-} mice indicating that the perturbation of plasmatic and platelet-mediated blood clotting observed *ex vivo* did not overtly influence the wound healing in this simple wounding model (data not shown).

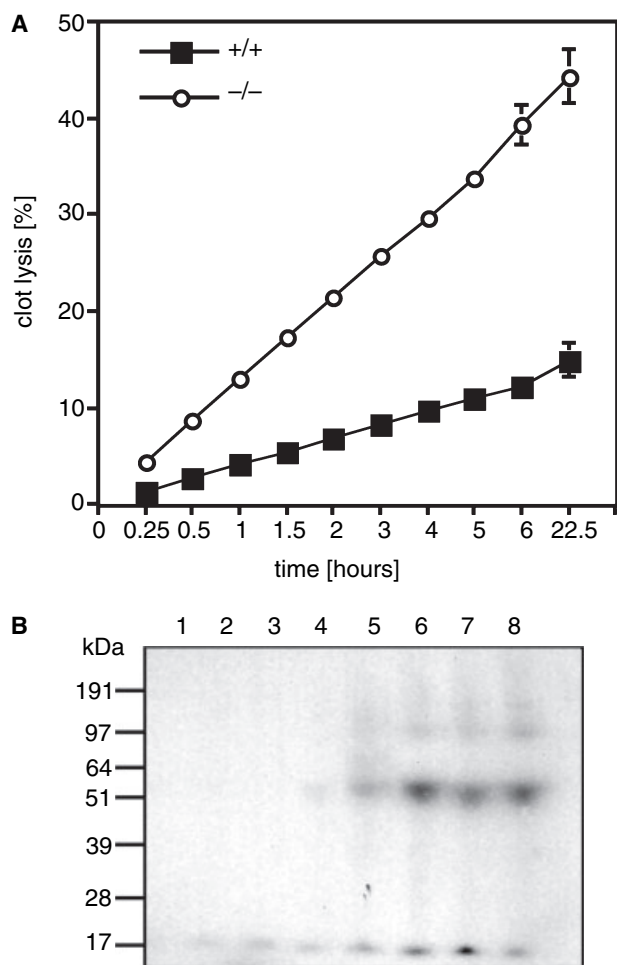


Fig. 7. Fibrinolytic activity in clotted mouse blood. Mouse blood was diluted with saline and clotted by the addition of Ca^{2+} and bovine thrombin using ^{125}I -labeled fibrinogen as a tracer. Fibrinolysis was measured as the release of ^{125}I radioactivity into the soluble clot supernatant. (A) Data represent the mean $\pm$ SEM of accumulated blood clot lysis of $Hrg^{+/+}$ (■, $n = 6$) and $Hrg^{-/-}$ (○, $n = 6$) mice expressed as percent of radioactivity of ^{125}I -labeled fibrinogen recovered in the lyzed clot supernatant. (B) Ten-microliter aliquots of lyzed clot supernatant harvested after 15, 30, 60 and 90 min of spontaneous clot lysis were separated on SDS-PAGE and autoradiographed. Lanes 1 and 5, 15 min; lanes 2 and 6, 30 min; lanes 3 and 7, 1 h; lane 4 and 8, 1.5 h. Note the abundance of fibrin degradation products in HRG-deficient clot supernatants (lanes 5–8) compared to wild-type clot supernatants (lanes 1–4).

Discussion

To analyze the physiological function of HRG we generated HRG-deficient mice by deleting the translation start point of exon 1 of the gene through homologous recombination. The *Hrg* null mutants had no immunologically detectable HRG protein in their plasma and the mice were fertile. We successfully bred homozygous HRG-deficient mice for several generations. These data indicate that HRG is dispensable during normal embryogenesis and development.

Homozygous HRG-deficient mice had significantly higher numbers of monocytes in their blood than wild-type mice. At present we have no explanation for this mild monocytosis.

Recently HRG was described as a potent opsonin [30] and we speculate that the lack of HRG could trigger monocyte proliferation to compensate for a potential decrease in phagocyte activation. A thorough investigation of the innate immune response in HRG-deficient mice will clarify the potential immune modulatory contribution of HRG.

A role of HRG in hemostasis has been repeatedly proposed based on the observation that HRG binds several components of the coagulation and fibrinolysis cascades. Prothrombin time was significantly shorter in $Hrg^{-/-}$ plasma samples than in $Hrg^{+/+}$ samples, and reconstitution of the plasma samples with purified human HRG normalized the prothrombin time. Hence we conclude that HRG may regulate the extrinsic pathway of coagulation in mice by interacting with thrombin. Our observation that HRG-deficient plasma also had higher antithrombin activity suggests an additional and alternative mechanism of HRG anticoagulant activity. Antithrombin is a potent physiological negative regulator of the blood coagulation inhibiting thrombin, factor Xa and factor IXa [31]. Its inhibitory activity towards serine proteases is much enhanced in the presence of heparin [32]. HRG has a high binding affinity for heparin and therefore potentially competes with the binding of heparin to antithrombin [33]. Consequently, HRG inhibits the high-affinity binding of antithrombin to thrombin as well as the antifactor Xa activity of antithrombin. Therefore, HRG is considered a modulator of anticoagulant activity of heparin in human plasma [34]. Comparing unmodified plasma samples of HRG-deficient mice and wild-type mice, we determined that HRG can antagonize antithrombin activity *in vivo*.

It is known that HRG interacts with fibrinogen *in vitro* [9]. HRG-deficient mice of both sexes had elevated plasma fibrinogen, especially female $Hrg^{-/-}$ mice, which had 35% more plasma fibrinogen than age- and sex-matched wild-type mice. This is the first demonstration *in vivo* that HRG levels in blood might also influence the level of the most abundant protein of the blood coagulation cascade, possibly associated with enhanced plasma coagulation. Similar observations exist in human patients. Three cases of congenital HRG deficiency have been reported to date [10–12]. Strikingly, all patients suffering from embolism were women. One patient with the HRG Tokushima 2 allele had only 50% of the normal HRG plasma level, and had associated elevated fibrinogen [12]. Given the shorter prothrombin time and the elevated fibrinogen levels one may ask, why a thrombophilic phenotype was not observed in HRG-deficient mice. One should recall at this point that HRG binds to plasminogen in plasma and is supposed to act as an antifibrinolytic [7]. Therefore the lack of HRG should be associated with faster fibrinolysis, which was indeed confirmed by measurements in $Hrg^{-/-}$ blood. This finding is compatible with a report that poly-histidine a surrogate of the histidine-rich region of HRG down-regulated fibrinolysis by suppressing plasmin interaction with fibrin [35]. Recently it was suggested that HRG mediated high-affinity tethering of plasminogen to cell surfaces, thus activating fibrinolysis [16]. This does not exclude the enhanced fibrinolysis observed in plasma of HRG-deficient mice, because the HRG

activity may vary greatly in free and cell-bound form as well as with local pH and Zn^{2+} concentration [14,17]. Regardless of the precise molecular mechanism our study strongly suggests that depending on the micro milieu HRG can act as both an anticoagulant and an antifibrinolytic *in vivo*.

Apart from plasmatic coagulation, hemostasis can also be modulated on the level of platelets, which are known to carry HRG [36]. To determine the effect of HRG deficiency on platelet function we measured the bleeding time following tail-tip amputation. Consistently, the bleeding time in *Hrg*^{-/-} mice was significantly shorter than in *Hrg*^{+/+} mice. This result suggested that in addition to its influence on plasmatic coagulation HRG inhibits platelet function. Concerning the possible molecular mechanism of the platelet inhibition, it was reported that HRG strongly binds platelets in the presence of Zn^{2+} [37], and inactivates platelet thrombospondin, a strong promoter of platelet aggregation [38]. As a result of its abundant histidine residues HRG is exquisitely sensitive to pH changes near the physiological blood pH and has been termed a pH sensor [15]. HRG may modulate both plasmatic and platelet-mediated hemostasis in a pH-dependent and Zn^{2+} -dependent fashion, matching specific requirements in different locations of the body that require enhanced or attenuated blood clotting. This hypothesis can now be tested using the HRG-deficient mice in various models of wounding and hemostasis.

Author contributions

N. Tsuchida-Straeten – cloning and sequencing mouse *Hrg* gene; construction of mouse *Hrg* targeting vector; DNA analysis of ES cell and mouse tail DNA with PCR and Southern blot analysis; measurement of antithrombin activity, prothrombin time, fibrinogen level, measurement of fibrinolytic activity; measurement of bleeding time. S. Ensslen – blastocyst injection of targeted ES cells; embryo transfer; genotyping and establishment of HRG-deficient mouse strains; skin wound experiments. C. Schäfer – genomic Southern blots; immunoblot analysis of mouse serum; maintenance of mouse colony. M. Wöltje – isolation and identification of genomic PAC clone carrying the mouse *Hrg* gene. B. Denecke – development of genomic PCR genotyping. M. Moser – electroporating ES cells; culture and transfection of ES cells; micromanipulation and embryo transfer. S. Gräber – immunoblot analysis; prothrombin time measurements; animal experimentation; vena cava bleeding for fibrinolytic analysis. S. Wakabayashi – *Hrg* cDNA cloning and provision; supervision of *Hrg* genomic and targeting vector subcloning. T. Koide – supervision and manuscript revision. W. Jahnke-Dechent – coordination of project and manuscript writing.

Acknowledgements

We thank Dr Fuminori Tokunaga for helpful discussion. Financial support was obtained from the START program of the Faculty of Medicine, RWTH Aachen (C.S.), the Japan Society for the Promotion of Science (T.K.), Academic

Frontiers Student Exchange Promotion Program in 2002–03 by Japanese Ministry of Education, Culture, Sports, Science and Technology (T.K.) and Deutsche Forschungsgemeinschaft (W.J.D.).

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